



EMMD: Question Bank on Unit-1

Q1. The runs scored by two batsmen in a seven-match T20I series are given below:

Match No.	1	2	3	4	5	6	7
Batsman-1	—	—	56	13	—	39	24
Batsman-2	20	10	5	47	29	4	12

Treat “—” as did not bat and ignore those entries. Using the coefficient of variation, decide which batsman is more consistent.

Answer: $CV_1 \approx 49.00\%$, $CV_2 \approx 78.80\%$; hence Batsman-1 is more consistent.

Q2. The number of customer-support tickets resolved by two employees over six working days is

Day	1	2	3	4	5	6
Employee A	18	20	22	19	21	20
Employee B	25	12	28	15	30	10

Calculate the coefficient of variation for each employee and determine who is more consistent.

Answer: $CV_A \approx 6.45\%$, $CV_B \approx 39.69\%$; hence Employee A is more consistent.

Q3. Prove that the standard deviation of the first n natural numbers $1, 2, \dots, n$ is

$$\sqrt{\frac{n^2 - 1}{12}}.$$

Answer: $\sigma = \sqrt{\frac{n^2 - 1}{12}}.$

Q4. The quiz marks (out of 20) of 40 students are summarized below:

Marks x	8	10	12	14	16	18	20
Frequency f	2	4	7	10	9	6	2

Calculate the arithmetic mean, variance, standard deviation, and coefficient of variation.

Answer: $\bar{x} = 14.3$, $\sigma^2 = 9.11$, $\sigma \approx 3.0183$, $CV \approx 21.11\%$.

Q5. For the frequency distribution

x	1	2	3	4	5
f	1	3	4	3	1

find:

- (i) the first four raw moments;
- (ii) the first four central moments;
- (iii) γ_1 , β_1 , β_2 , and γ_2 ;
- (iv) the nature of skewness and kurtosis.

Answer: (i) $\mu'_1 = 3$, $\mu'_2 = \frac{61}{6}$, $\mu'_3 = \frac{75}{2}$, $\mu'_4 = \frac{883}{6}$; (ii) $\mu_1 = 0$, $\mu_2 = \frac{7}{6}$, $\mu_3 = 0$, $\mu_4 = \frac{19}{6}$; (iii) $\gamma_1 = 0$, $\beta_1 = 0$, $\beta_2 = \frac{114}{49}$, $\gamma_2 = -\frac{33}{49}$; (iv) symmetric and platykurtic.

Q6. For the distribution

x	0	1	2	3	4
f	1	2	4	2	1

find the first four raw moments, the first four central moments, and the moment coefficients of skewness and kurtosis.

Answer: Raw moments: $\mu'_1 = 2$, $\mu'_2 = \frac{26}{5}$, $\mu'_3 = \frac{76}{5}$, $\mu'_4 = \frac{242}{5}$. Central moments: $\mu_1 = 0$, $\mu_2 = \frac{6}{5}$, $\mu_3 = 0$, $\mu_4 = \frac{18}{5}$. Skewness: $\gamma_1 = 0$, $\beta_1 = 0$. Kurtosis: $\beta_2 = \frac{5}{2}$, $\gamma_2 = -\frac{1}{2}$ (symmetric and platykurtic).

Q7. The first four moments of a distribution about 4 are 2, 20, 40, and 400, respectively.

- (i) Find the first four central moments.
- (ii) Find the coefficients of skewness and kurtosis.
- (iii) Find the arithmetic mean and standard deviation.
- (iv) Comment on skewness and kurtosis.

Answer: (i) $\mu_1 = 0$, $\mu_2 = 16$, $\mu_3 = -64$, $\mu_4 = 512$; (ii) $\gamma_1 = -1$, $\beta_1 = 1$, $\beta_2 = 2$, $\gamma_2 = -1$; (iii) $\bar{x} = 6$, $\sigma = 4$; (iv) negatively skewed and platykurtic.

Q8. The first four moments of a distribution about 2 are 1, 4, 16, and 64, respectively.

- (i) Find the first four central moments.
- (ii) Find the coefficients of skewness and kurtosis.
- (iii) Find the arithmetic mean and standard deviation.
- (iv) Comment on skewness and kurtosis.

Answer: (i) $\mu_1 = 0$, $\mu_2 = 3$, $\mu_3 = 6$, $\mu_4 = 21$; (ii) $\gamma_1 = \frac{2}{\sqrt{3}} \approx 1.1547$, $\beta_1 = \frac{4}{3}$, $\beta_2 = \frac{7}{3}$, $\gamma_2 = -\frac{2}{3}$; (iii) $\bar{x} = 3$, $\sigma = \sqrt{3} \approx 1.7321$; (iv) positively skewed and platykurtic.

Q9. For the observations

4, 5, 6, 7, 8, 9, 17,

calculate Karl Pearson's coefficient of skewness using the median and interpret the result.

Answer: $\bar{x} = 8$, median = 7, $\sigma = 4$, and $Sk = \frac{3(8-7)}{4} = 0.75$; the distribution is positively skewed.

Q10. The number of support requests received during nine hourly intervals is

4, 5, 5, 5, 6, 7, 8, 9, 14.

Find the arithmetic mean, mode, standard deviation, and Karl Pearson's coefficient of skewness. Interpret the sign of the coefficient.

Answer: $\bar{x} = 7$, mode = 5, $\sigma = \frac{\sqrt{76}}{3} \approx 2.9059$, and $Sk = \frac{3}{\sqrt{19}} \approx 0.6882$; the positive sign indicates positive skewness.

Q11. Find the Karl Pearson correlation coefficient between X and Y for the following data:

X	-1	1	2	4	6	7
Y	-1	2	3	3	5	8

Answer: $r = \frac{131}{\sqrt{19108}} \approx 0.9477$ (strong positive correlation).

Q12. An app ran ads over 6 days. The daily ad spends X , in thousand rupees, and new app downloads Y were recorded as follows:

Day	1	2	3	4	5	6
X	2	4	6	8	10	12
Y	30	50	65	80	95	110

Find the Karl Pearson correlation coefficient and interpret the result.

Answer: $r \approx 0.9986$; there is an extremely strong positive correlation between ad spend and downloads.

Q13. A coaching centre recorded the weekly practice hours X and test scores Y of six students:

X	2	3	4	5	6	7
Y	40	45	52	60	65	72

Find Karl Pearson's coefficient of correlation and interpret your answer.

Answer: $r \approx 0.9982$; there is an extremely strong positive correlation between practice hours and test scores.

Q14. For two correlated variables x and y , the regression line of y on x and that of x on y are

$$3x + 2y - 26 = 0$$

and

$$6x + y - 31 = 0,$$

respectively. Find:

- (i) $\bar{x}, \bar{y}$,
- (ii) the correlation coefficient r .

Answer: (i) $\bar{x} = 4, \bar{y} = 7$; (ii) $r = -\frac{1}{2}$.

Q15. For two correlated variables x and y , the regression lines are

$$8x - 10y + 66 = 0$$

and

$$14x - 18y + 124 = 0.$$

Find:

- (i) $\bar{x}, \bar{y}$,
- (ii) the correlation coefficient r between x and y .

Answer: (i) $\bar{x} = 13, \bar{y} = 17$; (ii) $r = \frac{\sqrt{35}}{6} \approx 0.9860$.

Q16. Fit a regression line of y on x for the following data:

x	1	2	3	4	5
y	2	4	5	4	6

Use the line to estimate y when $x = 6$.

Answer: Regression line: $y = 1.8 + 0.8x$; when $x = 6, \hat{y} = 6.6$.

Q17. Obtain the regression lines for the following data:

x	6	2	10	4	8
y	9	11	5	8	7

Also find:

- (i) the value of y when $x = 12$,
- (ii) the value of x when $y = 8$.

Answer: Regression lines: $y - 8 = -\frac{13}{20}(x - 6)$ and $x - 6 = -\frac{13}{10}(y - 8)$; (i) $\hat{y} = 4.1$ when $x = 12$; (ii) $\hat{x} = 6$ when $y = 8$.